W3a: the Smith walk via the elliptic 2-isogeny class

Frozen-protocol run · protocol frozen at `eba0a48fbf` · runs 2026-10-26 UTC (run-20261026T0212Z, run-20261026T0244Z)

Synthetic curves over $\mathbb{F}_{p^3}$ only. No claim about any standardized curve, no sub-rho or key-recovery claim.

1. Question

The problem record `problems/smith-kernel-isogeny-class-walk.md` asks what an isogeny walk costs when it moves a genus-3 hyperelliptic curve H to a member of its class where Smith's $(2,2,2)$-construction succeeds. Hyperelliptic-to-hyperelliptic $(2,2,2)$-isogenies generically do not exist (see `LITERATURE.md`). W3a therefore walks on the elliptic side.

- Start from E/K, $K = \mathbb{F}_{p^3}$, admitting a Joux–Vitse / Momose–Chao $(m, r) = (3, 5)$ descent to $H/\mathbb{F}_p$.
- Move E by rational 2-isogenies (Vélu).
- Re-descend, and run Smith's full test (`smith_prop24::analyze_curve`, `full_success`).

Q1 is exhaustive at small p. Q2 is sampled walks from $p \approx 2^{10}$ to 2^{20}, priced in $\mathbb{F}_p$ multiplications. Q3, the literature, is in `LITERATURE.md`.

2. Q1: exhaustive at $p \in \{7, 11, 13, 17, 19, 23, 29, 31\}$ (measured)

Every reduced weak model ($2p^2$ per p) was descended and tested. Every class was closed under rational 2- and 3-isogenies. All controls had zero failures at every p:

- the L-polynomial of every H;
- trace equality and a dual edge on every edge;
- JV-model signatures;
- no `analyze_curve` failure.

The unreduced $p = 7$ enumeration (98,784 models) gives the same classes and signature sets. A lost 2026-10-20 run of the same cells printed identical summaries.

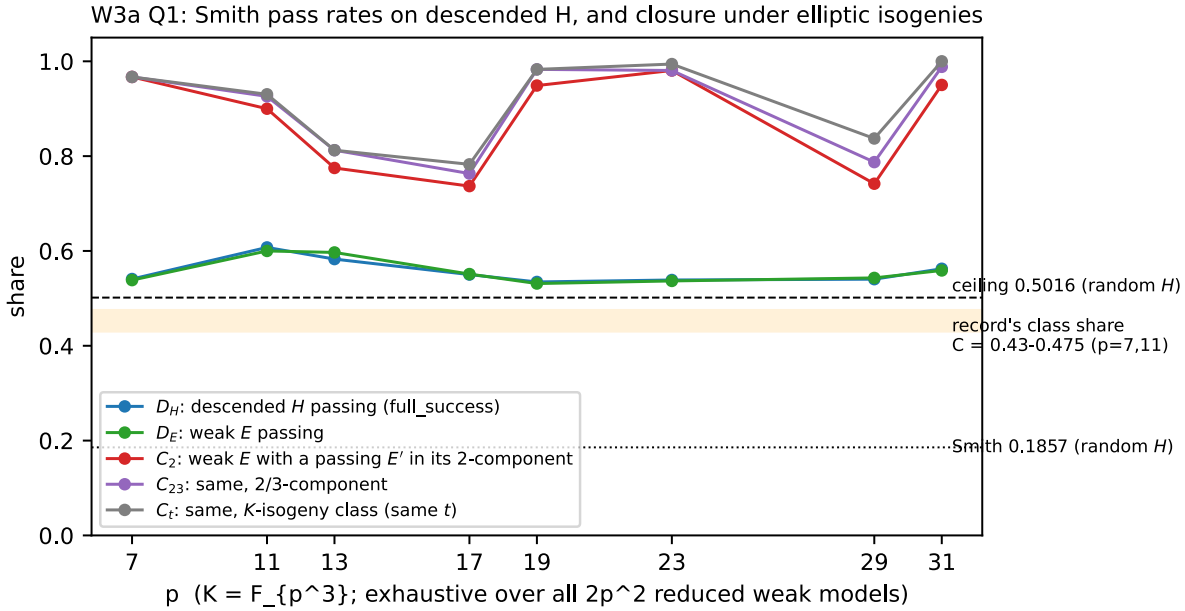


Figure 1: Shares against p: D_H (descended H passing), D_E (weak E passing), C_2, C_{23}, C_t (weak E whose 2-component, 2/3-component or same-trace class holds a passing E). The reference lines are Smith's 0.1857 and 0.5016 for random H, and the record's class share 0.43–0.475 at $p = 7, 11$. Populations: 91 to 1,886 weak E per p; exhaustive, so there is no sampling error.

Findings.

1. **Every descended H has a rational tractable subgroup** (share 1.000). Only two cycle types occur, 1+1+3+3 and 2+6.
   - Prop. 2 holds for 0.78–0.88 of H, and Prop. 4 given Prop. 2 for 0.62–0.72.
   - D_H = 0.53–0.61, which is 3× Smith's 0.1857 for random curves.
   - Pre-registered E3 is falsified: it is a structural departure of the descended locus, not a bug.
   - The record's ceiling (0.43–0.50, measured on random H) does not apply to this locus.
2. **Closure is large.**
   - C_2 = 0.74–0.98, C_{23} = 0.76–0.99, C_t = 0.78–1.00.
   - r_2 = 0.68–1.00 and r_{23} = 0.83–1.00.

- Passing is not constant on components: 38–75% of the components with at least 2 weak curves are mixed. E2 holds, and the walk is not killed by invariance.

3. **Distances are short** (Figure 2). A failing weak E is a mean 1.5–3.6 rational 2-isogenies from the nearest passing weak E, with a maximum of 3–11. With 3-isogenies added it is 1.4–2.9.
4. **The residue of p mod 4 matters.** The share of failing E with no passing E in their component is 56–59% at $p = 13, 17, 29$, against 4–25% at $p \equiv 3 \pmod 4$.

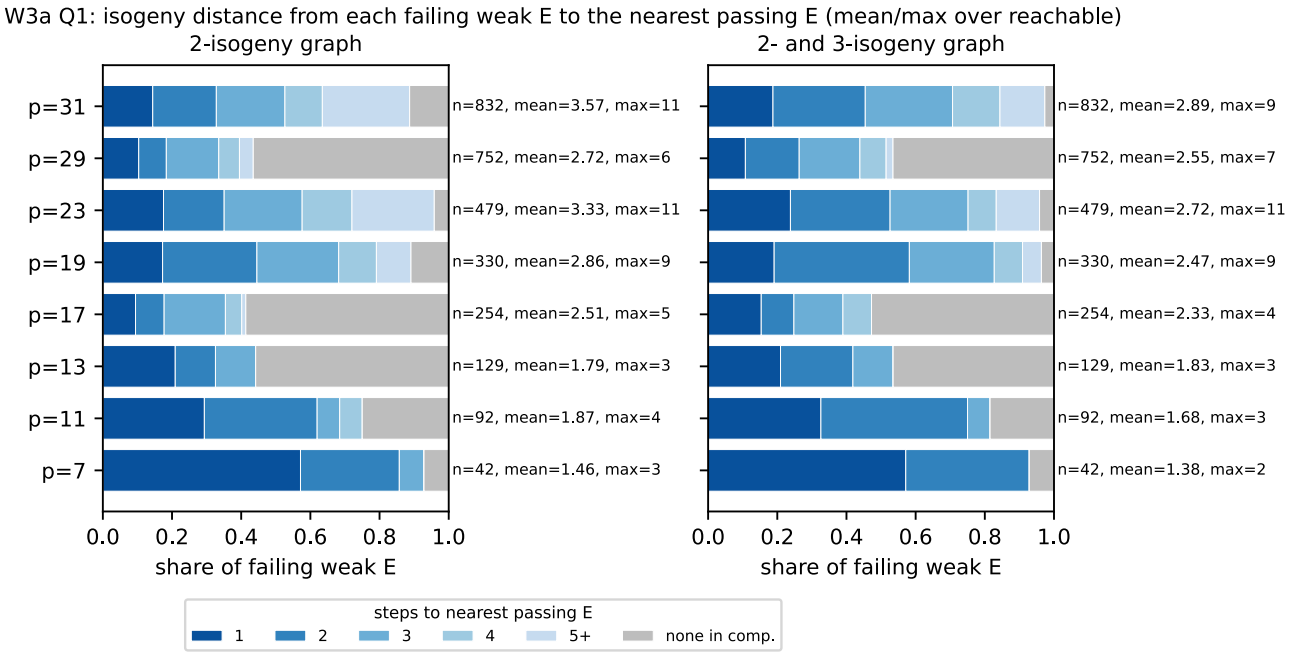


Figure 2: Distance from each failing weak E to the nearest passing weak E, as shares of the failing E. The n, mean and max on the right are over reachable ones. Measured, exhaustive.

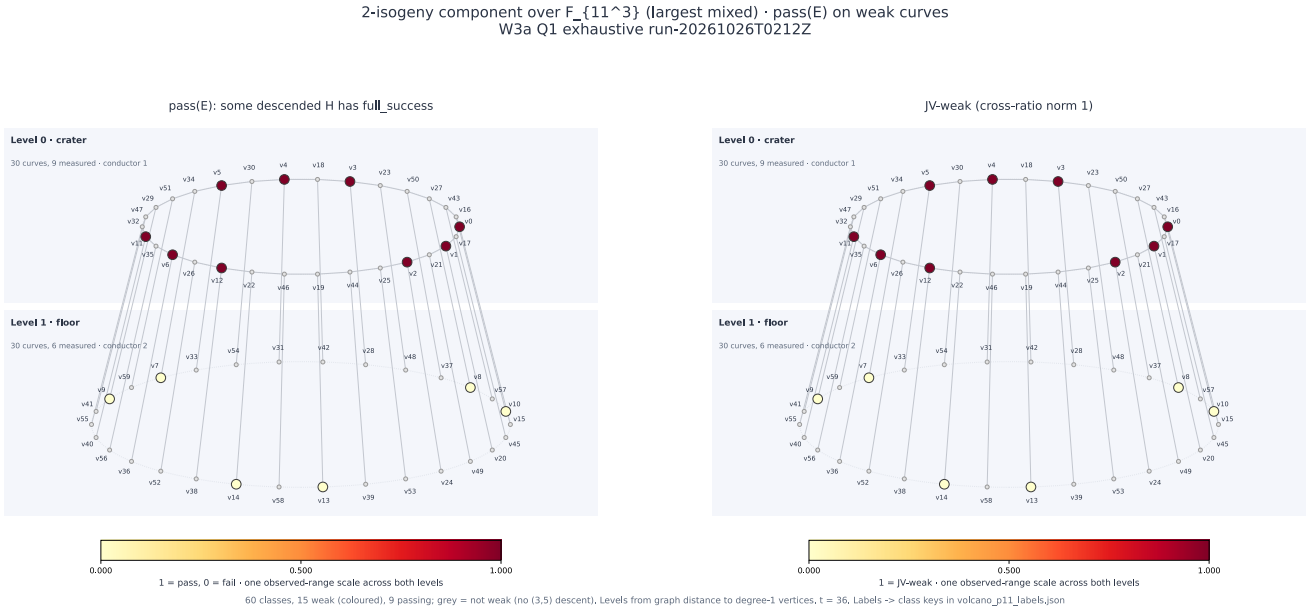


Figure 3: The largest mixed 2-isogeny component at $p = 11$ (60 classes, 15 weak, 9 passing; $t = 36$), drawn with `tools/volcano-diagram`. Levels come from the graph distance to degree-1 vertices. The label to class-key map is in `figures/volcano_p11_labels.json`. Here the weak crater curves are JV-weak and pass, and the weak floor curves fail.

3. Q2: sampled walks (measured; Smith cost derived)

Frozen primes are the first prime above 2^k, $k = 10..16, 18, 20$. There are 40, 24, 12 or 6 walks per arm and prime: 500 walks in all, from 2026-10-26T02:44 to 04:01 UTC.

- **Arm A**: from a random failing JV-split weak E, a non-backtracking 2-walk through full-2-torsion nodes until a JV-weak node's model passes.
- **Arm B**: from a random full-2-torsion curve until the first JV-weak node.

Caps: $2 \cdot 10^7$ tested nodes and 3,300 s per walk, plus the amended step cap and trap rule (Section 4).

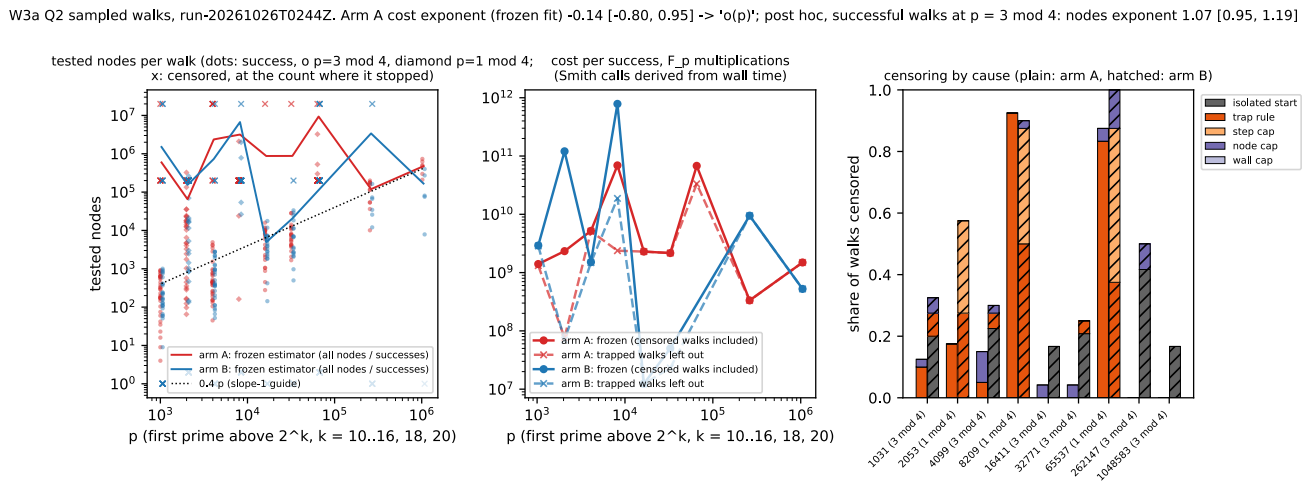


Figure 4: Left: tested nodes per walk. Successes are dots, censored walks are crosses. Lines show the frozen estimator (all tested nodes over successes). Middle: cost per success in $\mathbb{F}_p$ multiplications (frozen, and with trapped walks left out). Right: censoring share by cause. Data: `data/walks/walk_*.jsonl`, `fit.json`.

Unit costs (`fit.json`; multiplications are instrumented):

- one step: 1.8–3.0 k $\mathbb{F}_p$ multiplications, growing like $\log p$ because of the square root in K;
- one JV-weak test: 114;
- one descent: 2.0–4.0 k;
- one Smith call: 25–61 k (derived from wall time at 5.1 ns per multiplication; inversion = 7.6 multiplications; shared host).

The parity of admissible steps [derived, and tested by `admissible_forward_steps_follow_the_sign_of_minus_one`]. For the roots r_0, r_1, r_2 of a node, $\prod_i (r_i - r_j)(r_i - r_l) = -\prod_{i<j} (r_i - r_j)^2$. So the number of squares among the three is

- 2 or 0 when $\chi(-1) = -1$, that is $p \equiv 3 \pmod 4$. Every non-start node then has exactly one forward edge (a non-branching path), and a start can be isolated.
- 1 or 3 when $p \equiv 1 \pmod 4$, so each node is a leaf or branches into three.

The data agree. At $p \equiv 3 \pmod 4$ there are no reflections, and arm A never revisits a class on a successful walk. At $p = 2053, 8209, 65537$, arm A walks are trapped in closed sets of 6–30 classes: 7/40, 37/40 and 20/24 walks. Arm B has 0/24 successes at 65537.

Scaling.

- **The frozen rule outputs `o(p)`.** The arm A cost exponent is -0.14 $[-0.80, 0.95]$, with primes and walks resampled.
- **We do not read that output as evidence.** In the frozen estimator, a censored walk contributes what it had accrued when a cap stopped it. Up to $2 \cdot 10^7$ nodes per walk dominate the means at the small primes. At $p = 1031$, for example, the mean is $5.95 \cdot 10^5$ nodes while the median successful walk is 291. This flattens the slope.
- **Post hoc, labelled:** for successful walks at $p \equiv 3 \pmod 4$:
  - tested nodes scale with exponent 1.07 $[0.95, 1.19]$, and cost with 1.13 $[1.00, 1.26]$. The rule's label for the cost is `omega(p)`; with the $\log p$ growth of the step cost this is $p \log p$, that is $\tilde{\Theta}(p)$;
  - medians are 0.1–$0.6p$;
  - $h(p) \approx (3 \text{ to } 10)/p$;
  - at $p = 262147$ and 1048583, $\pi = 0.80$ (12/15 hits) and 0.55 (6/11 hits). The second is consistent with Q1′s pass given JV (0.78–0.85) within sampling at 11 hits;
  - a non-circular check (derived): the walk's h with Q1′s $\pi \approx 0.8$ predicts $1/(0.8h) = 1.19 \cdot 10^5$ and $3.24 \cdot 10^5$ nodes at $p = 262147$ and 1048583, against $1.19 \cdot 10^5$ and $4.75 \cdot 10^5$ observed (ratios 1.00 and 0.68).
- Arm B, successful walks at $p \equiv 3 \pmod 4$: exponent 0.93 $[0.75, 1.10]$.

Verdict. The walk costs $\Theta(p)$ nodes at about 2.5 k multiplications each. That is not $o(p)$, so it does not vanish against $\tilde{O}(q)$ index calculus. At $p \equiv 1 \pmod 4$ the walk as specified usually never finishes. E1 is supported.

4. Amendments and deviations

`PROTOCOL.md` is unchanged. The full list is in `README.md`. In brief:

1. A step cap of 4 × the node cap, for reflection loops that never retest a node.
2. An isolated start (no full-2-torsion neighbour) is censored. Before this, the arm B pilot panicked: a control failure, fixed before data were collected.

3. A distinct-j diagnostic, outside the cost counters, and a trap rule: a walk is censored once it has tested $\geq 2 \cdot 10^5$ nodes with ≤ 1000 distinct j.
4. The fit records censoring by cause. It bootstraps over primes and then walks, as frozen; the old driver resampled walks only. A prime with no success gives "undetermined". Post hoc variants are labelled.
5. Host: an Apple M4 Pro with 14 cores, shared and without pinning (cores 29 and 30 do not exist), running ≤ 4 processes at nice 10. Wall time and the derived Smith cost are noisy.
6. Calibration: the $p = 65537$ row prices every p.
7. "Nodes visited" means tested nodes, counting retests.
8. Q1 was extended from the brief's $\{7, 11, 13\}$ to the protocol's full list.
9. All data come from the 2026-10-25/26 rerun. The 2026-10-20 outputs were lost.

5. What this establishes and what not

Established on this locus and range:

- the pass rates;
- isogeny closure and distances (exhaustive, $p \leq 31$);
- per-operation costs;
- $\Theta(p)$ walk length for the walks that finish (post hoc, at six primes $\equiv 3 \pmod 4$ up to 2^{20});
- frequent trapping at $p \equiv 1 \pmod 4$.

Not established:

- whether some other walk policy avoids the traps, for example admitting steps through non-full-2-torsion nodes;
- whether weak or passing curves cluster beyond density $1/p$, which is what a cheaper walk would need;
- why the descended locus passes Smith at ~ 0.55;
- any statement beyond $p \approx 2^{20}$.

Next step: measure clustering of weak and passing curves in the isogeny graph. General $(3, 5)$-weak curves are sparse too: $2p^2$ models against about $2p^3$ K-classes, so their density is about $1/p$. At that density the nearest weak node is about $\log_2 p$ steps away, and a ball of that radius holds about p nodes. A general weakness test plus a short search would therefore beat $\Theta(p)$ only if weak or passing curves cluster beyond $1/p$.

The Q1 data give a first, derived reading. The weak share of the explored components falls from 0.54 at $p = 7$ to 0.13 at $p = 31$. The global share is $91/686 \approx 0.13$ and $1886/59582 \approx 0.03$ (with $2p^3$ classes), so the explored components are about 4× enriched at $p = 31$. The mean distance rises from 1.5 to 3.6.

The discriminating step is to test whether that enrichment persists as p grows, and whether d_2 grows like $\log p$, on larger exhaustive graphs or sampled components. Only then is a general weakness test worth building.